

Git and GitHub Installation Guide for Beginners

Windows 10/11 • Step-by-step installation, setup, and first upload

Purpose: help a complete beginner install Git, create a GitHub account, connect a local project, and upload it safely.

1. What are Git and GitHub?

Git is a version control system installed on your computer. It records changes to files, lets you create commits, compare versions, create branches, and return to earlier versions.

GitHub is an online platform that hosts Git repositories. It helps you store code online, back up your work, share projects, and collaborate with other developers.

Simple difference:

- Git = the tool that tracks changes locally.
- GitHub = the online platform where Git repositories can be stored and shared.

2. What you need before starting

- A Windows 10 or Windows 11 computer.
- Internet connection.
- A GitHub account.
- A project folder or a practice folder.
- A browser such as Chrome or Edge.

Important: Git and GitHub are not the same application. You install Git on Windows, and you create or use your GitHub account in a browser.

3. Download Git for Windows

1. Open your browser.
2. Go to the official Git for Windows page: <https://git-scm.com/install/windows>
3. Under Standalone Installer, choose “Git for Windows/x64 Setup” for most Intel or AMD Windows computers.
4. Choose ARM64 Setup only if your Windows computer uses an ARM processor.
5. Do not choose Portable unless you specifically need a version that runs without normal installation.

4. Install Git

1. Open the downloaded .exe installer.
2. If Windows asks for permission, select Yes.
3. Click Next through the setup screens.
4. Keep the default installation folder unless you have a specific reason to change it.
5. Keep the recommended default components.
6. For the editor, you may keep the default or select Visual Studio Code if you already use it.
7. Continue with the recommended settings.
8. Click Install.
9. When installation finishes, click Finish.

For a beginner, the default options are normally sufficient.

5. Verify the installation

1. Open the Start Menu.
2. Search for Git Bash and open it.
3. Run:

```
git --version
```

If a version number appears, Git is installed correctly.

You can also open PowerShell and run the same command.

6. Create or sign in to GitHub

1. Open <https://github.com/>
2. Select Sign up to create an account, or Sign in if you already have one.
3. Complete the email verification and account setup steps.
4. Do not show your password, recovery codes, or access tokens in a screen recording.

7. Configure Git for the first time

Open Git Bash and run these commands once on your computer. Replace the example values with your own name and email.

```
git config --global user.name "Your Name"
git config --global user.email "your-email@example.com"
```

To check the saved configuration:

```
git config --global --list
```

The name and email are used as the author information for your commits.

8. Create a GitHub repository

1. Sign in to GitHub.
2. Select the + menu in the upper-right corner.
3. Select New repository.
4. Enter a repository name, for example: git-demo
5. Add an optional description.
6. Choose Public or Private.
7. If you are uploading an existing local project, leave README, .gitignore, and license options unchecked for the simplest first push.
8. Select Create repository.
9. Copy the repository HTTPS URL.

9. Upload an existing local project using Git Bash

Open Git Bash inside your project folder. You can right-click inside the folder and choose Open Git Bash here if that option is available.

Run these commands in order:

```
git init
git add .
git commit -m "Initial commit"
git branch -M main
git remote add origin https://github.com/YOUR-USERNAME/YOUR-REPOSITORY.git
git push -u origin main
```

Replace the repository URL with your own URL.

What each command does:

- `git init` — creates or initializes the local Git repository.
- `git add .` — stages the files in the current folder.
- `git commit` — saves a snapshot of the staged files.
- `git branch -M main` — names the current branch main.
- `git remote add origin` — connects the local project to GitHub.
- `git push -u origin main` — uploads the main branch to GitHub.

10. Safer alternative when origin already exists

If you see:

error: remote origin already exists.

Do not add origin again. Update the existing URL:

```
git remote set-url origin https://github.com/YOUR-USERNAME/YOUR-REPOSITORY.git
```

Check it:

```
git remote -v
```

Then push:

```
git push -u origin main
```

11. Common errors and fixes

Error: pathspec 'README.md' did not match any files

Cause: README.md does not exist in the current folder.

Fix: Use the real filename, for example:

```
git add index.txt
```

Or create a README.md file first.

Error: nothing added to commit but untracked files present

Cause: You tried to commit before staging files.

Fix:

```
git add .  
git commit -m "Initial commit"
```

Error: src refspec main does not match any

Cause: There is no successful commit on the main branch.

Fix: Stage and commit first, then push.

Error: remote origin already exists

Cause: The remote is already configured.

Fix:

```
git remote -v  
git remote set-url origin YOUR-REPOSITORY-URL
```

Error: rejected because the remote contains work

Cause: GitHub already has commits that are not in your local repository.

Fix: Do not use `--force`. Inspect the repository and synchronize carefully.

12. Everyday Git workflow

After editing files in your project, use:

```
git status
git add .
git commit -m "Describe your change"
git push
```

To download changes made on GitHub:

```
git pull
```

To copy an existing GitHub repository to your computer:

```
git clone https://github.com/OWNER/REPOSITORY.git
```

13. Final checklist

- Git installed successfully.
- `git --version` shows a version.
- GitHub account created or opened.
- Git username and email configured.
- GitHub repository created.
- Local project initialized.
- Files staged and committed.
- Remote origin connected.
- Project pushed to GitHub.
- Repository refreshed and files visible online.

14. Official references

Git for Windows installation:

<https://git-scm.com/install/windows>

Git first-time setup:

<https://git-scm.com/book/en/v2/Getting-Started-First-Time-Git-Setup>

GitHub: Set up Git:

<https://docs.github.com/en/get-started/git-basics/set-up-git>

GitHub: Create a repository:

<https://docs.github.com/en/repositories/creating-and-managing-repositories/creating-a-new-repository>

GitHub: About Git:

<https://docs.github.com/en/get-started/using-git/about-git>

Note: GitHub interface labels can change slightly over time. If a button has moved, use GitHub's official documentation linked above.